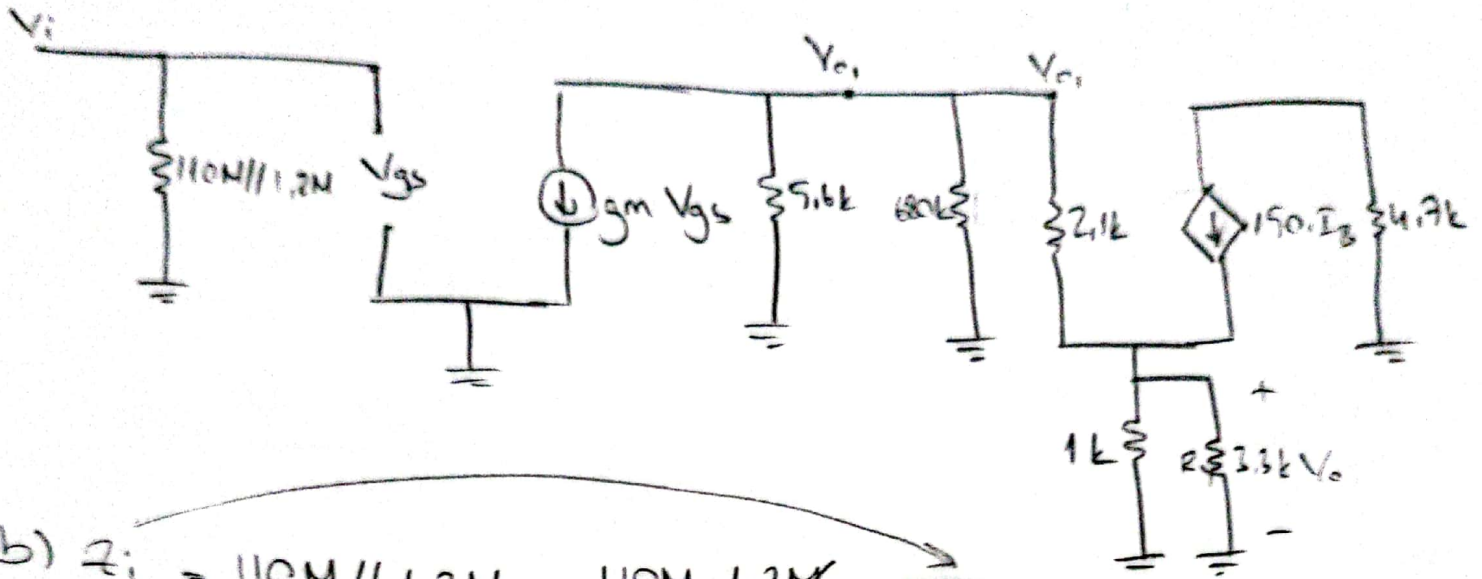


[Signature]

Soru 1;

a)



$$b) Z_{i1} = 110M // 1.2M = \frac{110M \cdot 1.2M}{1.2M} = 1.18M \Omega$$

$$Z_{i2} = (5.6k // 680k) // Z_B \quad Z_B = \frac{V_{c1}}{I_B}$$

$$Z_{i2} = \frac{(5.6k // 680k)}{5.5k} // 116.8k$$

$$Z_{i2} = 5.29k \Omega$$

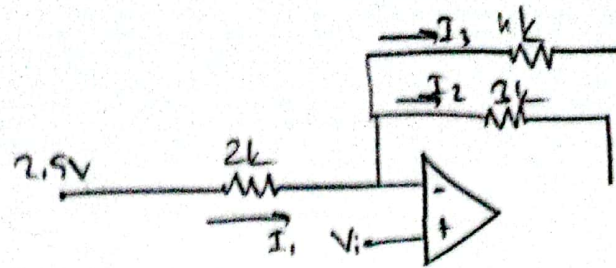
$$Z_o = Z_{i2} (3.3k \text{ dahil değil})$$

$$c) A_v = -g_m (R_D // R_Y)$$

$$A_{v1} = (-3 \times 10^{-3}) \cdot (5.6k // (680k // Z_B))$$

$$A_{v2} = (-3 \times 10^{-3}) \cdot (5.29k // 3.3k)$$

Soru 2;



$$\frac{2.5 - V_i}{2k}$$

$$V_{+2} = \frac{V_{o1}}{4k + 2k} \cdot 2k = \frac{V_{o1}}{3}$$

$$I_1 = I_2 + I_3$$

$$I_3 = \frac{V_i - V_{o2}}{6k}$$

$$\frac{2.6V_i - 3}{3}$$

$$\frac{2.5 - V_i}{2k} = \frac{V_i - V_{o1}}{3k} + \frac{V_i - (V_{o1}/3)}{4k}$$

$$15 - 6V_i = 4V_i - 4V_{o1} + 3V_i - V_{o1}$$

$$4V_{o1} + V_{o1} = 13V_i - 15$$

$$5V_{o1} = 13V_i - 15$$

$$V_{o1} = \frac{13V_i - 15}{5}$$

$$V_{o1} = 2.6V_i - 3$$

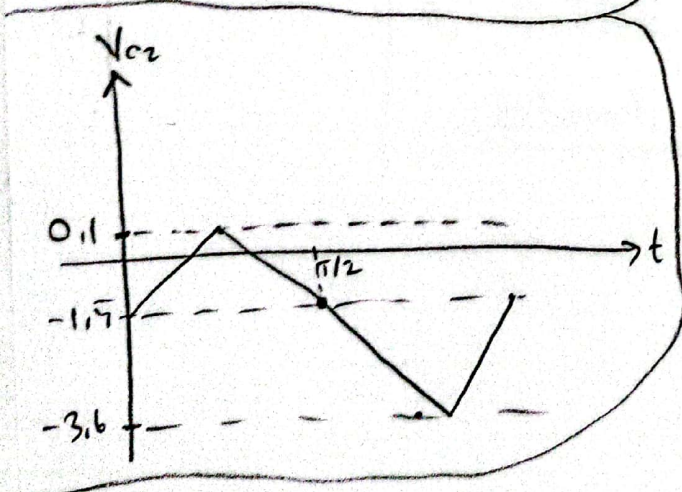
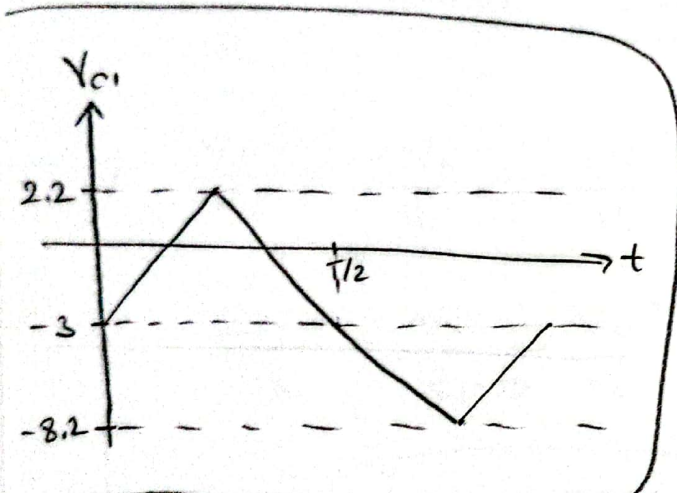
$$\frac{V_{o1}}{3} - V_{o2} = \frac{V_i - V_{o2}}{6k}$$

$$V_{o1} - 3V_{o2} = V_i - V_{o2}$$

$$2.6V_i - 3 - 3V_{o2} = V_i - V_{o2}$$

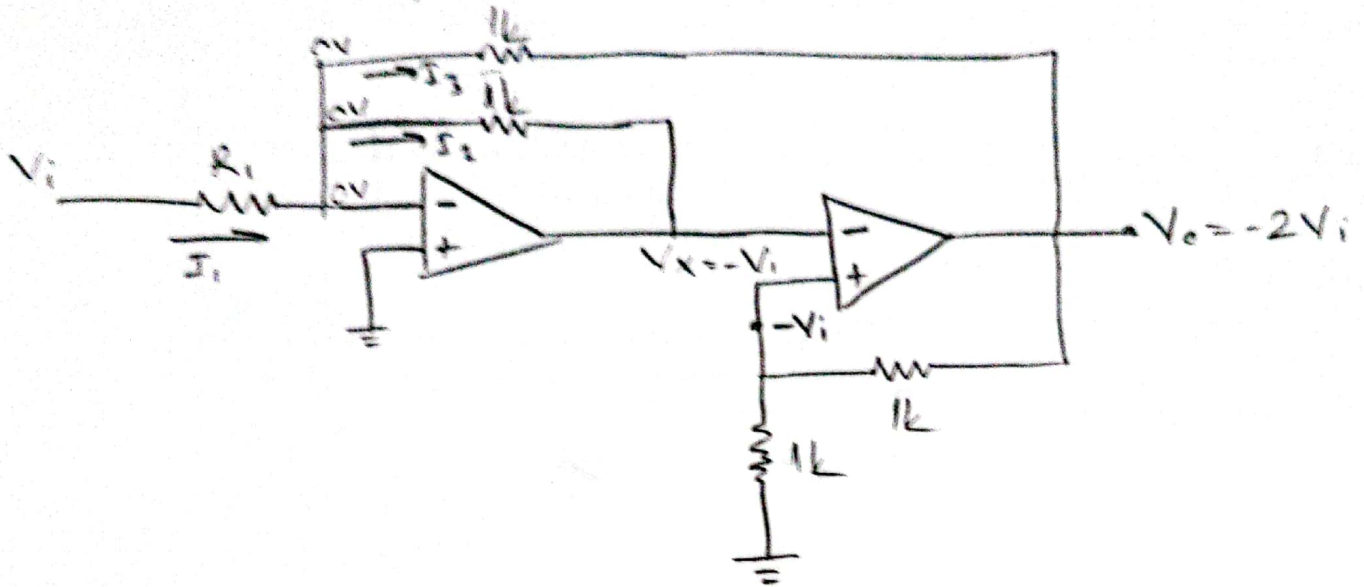
$$2V_{o2} = 1.6V_i - 3$$

$$V_{o2} = 0.8V_i - 1.5$$



Ölçü

Soru 3; $A_v = \frac{V_o}{V_i} = -2$ $V_o = -2V_i$



$$\frac{V_i - 0}{R_1} = \frac{0 - V_x}{1k} + \frac{0 - (-2V_i)}{1k}$$

$$\frac{V_i}{R_1} = \frac{-V_x}{1k} + \frac{2V_i}{1k}$$

$$V_x = -V_i \text{ olmalı}$$

$$\frac{2V_i}{1k} - \frac{V_i}{R_1} = \frac{V_x}{1k} \quad V_x = -V_i$$

$$\frac{2V_i}{1k} - \frac{V_i}{R_1} = \frac{-V_i}{1k} \text{ olmalı}$$

$$\boxed{R_1 = 0,33k \text{ olmalı}}$$

$$R_1 = \frac{1}{3} k$$

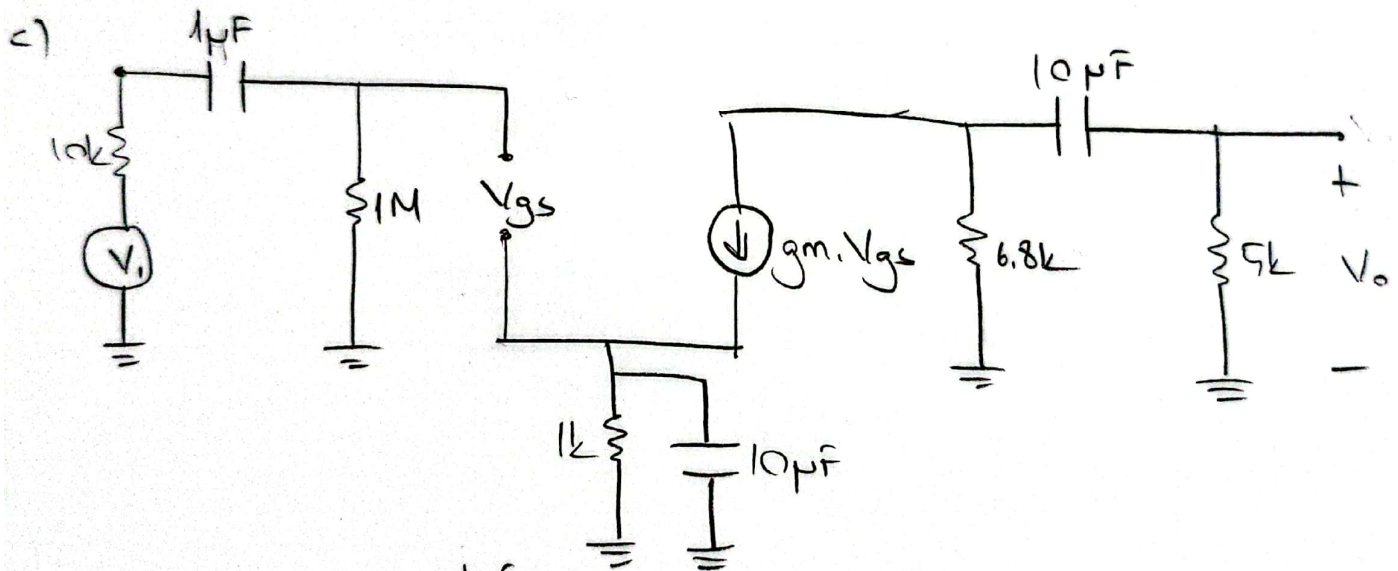
Soru 4;

$$\frac{6.8k \cdot 5k}{11.8k}$$

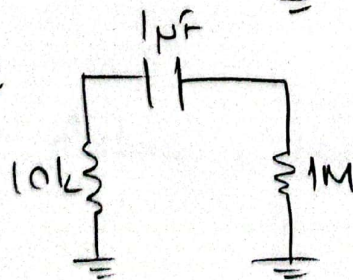
a) $z_i = 1M$, $z_o = 6.8k$

b) $A_v = \frac{V_o}{V_i} = \frac{(-g_m \cdot V_{gs}) \cdot (6.8k \parallel 5k)}{V_{gs}} = (2 \times 10^{-3}) \cdot (2.88k)$

$A_i = \frac{-A_v \cdot z_i}{R_y} = \frac{(-5.76) \cdot \frac{200}{1 \times 10^6}}{5 \times 10^{-5}} = \frac{-5.76 V}{5 \times 10^{-5}} = -1.15k A = A_i$



* 1pF'in etkisi;

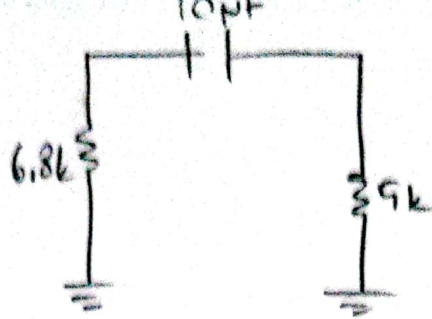


$$f_1 = \frac{1}{2\pi \cdot 1pF \cdot (10k + 1M)}$$

$$\frac{1}{2\pi \cdot (1.01)}$$

$$= \frac{1}{6.34} = 0.15 Hz$$

* 10pF 'in etkisi;



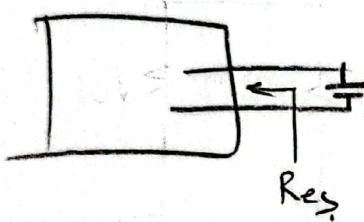
$$f_2 = \frac{1}{2\pi \cdot 10^{-3} \cdot 11.8}$$

$$\frac{1}{2\pi \cdot 11.8 \cdot 10^{-3}}$$

$$\frac{1000}{741.04}$$

$$= 1.34\text{Hz}$$

* $(10\text{pF})_2$ 'nin etkisi;



$$\frac{1}{2\pi \cdot (10\text{pF}) \cdot R_{es}} \rightarrow 476.13$$

$$\frac{1}{\frac{29304.7}{10^6}}$$

$$R_{es} = R_s \parallel \frac{1}{\frac{2 \times 10^{-3}}{500}}$$

$$= \frac{10000 \cdot 500}{10500}$$

Alt kesim frekansı = $\boxed{33.43\text{Hz}}$ $f_3 = 33.43\text{Hz}$

Soru h devamı